Year 11 Foundation

REVISION PACK

Graphs 5

Name: _____

Class: _____

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: adding the equations

Solve the simultaneous equations

$x + y = 9$

$x - y = 3$

Step 1	$+y$ and $-y$: add	$2x = 12$
Step 2	Divide by 2	$x = 6$
Step 3	Put x in	$6 + y = 9$
Step 4	Subtract 6	$y = 3$

Practice: adding the equations

1.1 Solve the simultaneous equations

$x + y = 12$

$x - y = 4$

1.2 Solve the simultaneous equations

$x + y = 11$

$x - y = 7$

1.3 Solve the simultaneous equations

$x + y = 12$

$x - y = 2$

1.4 Solve the simultaneous equations

$x + y = 13$

$x - y = 7$

1.5 Solve the simultaneous equations

$x + y = 7$

$x - y = 5$

1.6 Solve the simultaneous equations

$x + y = 15$

$x - y = 7$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: subtracting the equations

Solve the simultaneous equations

$4x + y = 17$

$x + y = 8$

Step 1	$+y$ and $+y$: subtract	$3x = 9$
Step 2	Divide by 3	$x = 3$
Step 3	Put x in	$3 + y = 8$
Step 4	Subtract 3	$y = 5$

Practice: subtracting the equations

1.7 Solve the simultaneous equations

$2x + y = 13$

$x + y = 8$

1.8 Solve the simultaneous equations

$4x + y = 14$

$x + y = 5$

1.9 Solve the simultaneous equations

$3x + y = 19$

$x + y = 7$

1.10 Solve the simultaneous equations

$5x + y = 17$

$x + y = 9$

1.11 Solve the simultaneous equations

$2x + y = 20$

$x + y = 12$

1.12 Solve the simultaneous equations

$3x + y = 21$

$x + y = 13$

Graphs 5 · Simultaneous equations (Lesson 3.1)

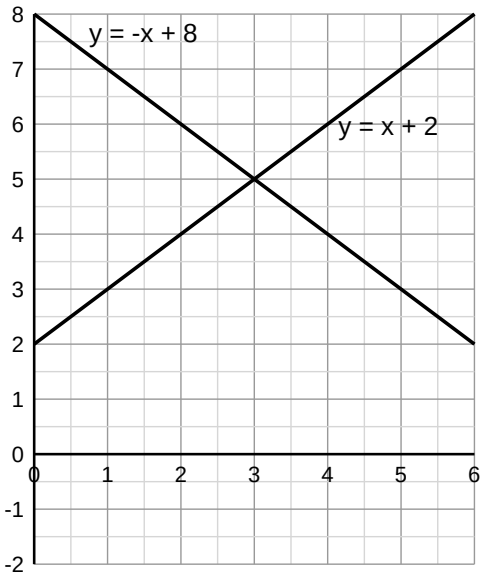
Worked example: solving using graphs

The graphs of $y = x + 2$ and $y = 6 - x$ are drawn on a grid. They cross at (2, 4). Use the graphs to solve the simultaneous equations.

Step 1	Where the lines cross	$(2, 4)$
Step 2	Answer	$x = 2,\ y = 4$

Practice: solving using graphs

1.13



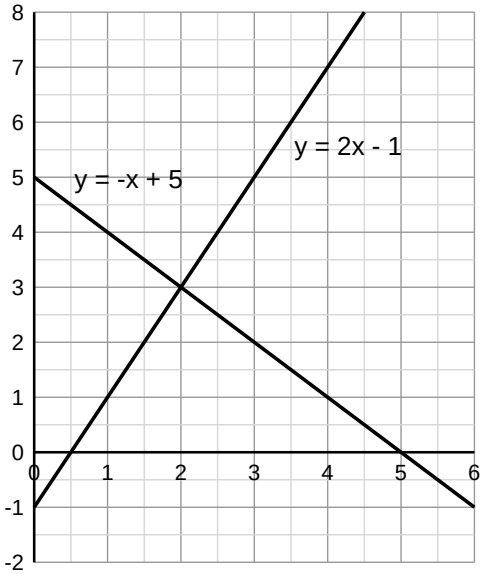
The graphs of $y = x + 2$ and $y = -x + 8$ are drawn on the grid.

Use the graphs to solve the simultaneous equations

$y = x + 2$

$y = -x + 8$

1.14



The graphs of $y = 2x - 1$ and $y = -x + 5$ are drawn on the grid.

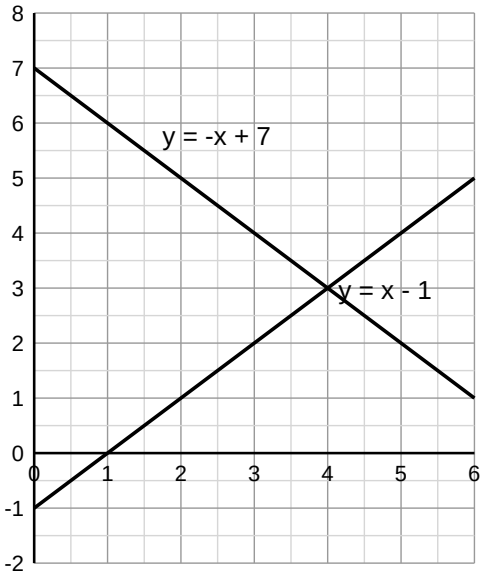
Use the graphs to solve the simultaneous equations

$y = 2x - 1$

$y = -x + 5$

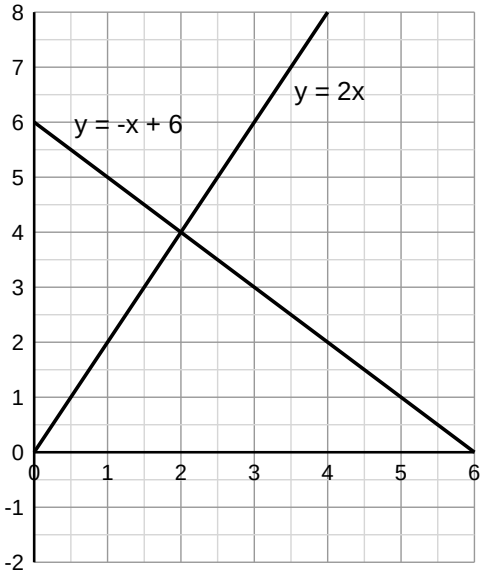
Practice: solving using graphs (continued)

1.15



The graphs of $y = x - 1$ and $y = -x + 7$ are drawn on the grid.
Use the graphs to solve the simultaneous equations
$y = x - 1$
$y = -x + 7$

1.16



The graphs of $y = 2x$ and $y = -x + 6$ are drawn on the grid.
Use the graphs to solve the simultaneous equations
$y = 2x$
$y = -x + 6$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: adding with a number in front of x

Solve the simultaneous equations

$2x + y = 9$

$x - y = 3$

Step 1	$+y$ and $-y$: add	$3x = 12$
Step 2	Divide by 3	$x = 4$
Step 3	Put x in	$4 - y = 3$
Step 4	Add y, subtract 3	$y = 1$

Practice: adding with a number in front of x

1.17 Solve the simultaneous equations

$2x + y = 12$

$x - y = 3$

1.18 Solve the simultaneous equations

$4x + y = 13$

$x - y = 2$

1.19 Solve the simultaneous equations

$3x + y = 11$

$x - y = -3$

1.20 Solve the simultaneous equations

$5x + y = 11$

$x - y = 1$

1.21 Solve the simultaneous equations

$2x + y = 15$

$x - y = 3$

1.22 Solve the simultaneous equations

$4x + y = 10$

$x - y = -5$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: equations with no solution

Explain why the simultaneous equations $y = 2x + 3$ and $y = 2x - 1$ have no solution.

Step 1	Look at the gradients	Both lines have gradient 2
Step 2	Same gradient, different intercepts	The lines are parallel
Step 3	Answer	Parallel lines never meet, so there is no point that fits both equations

Practice: equations with no solution

2.1 Do the simultaneous equations $y = 3x + 1$ and $y = 3x - 4$ have a solution?
Give a reason for your answer.

2.2 Do the simultaneous equations $y = 4x + 2$ and $y = -x + 7$ have a solution?
Give a reason for your answer.

2.3 Do the simultaneous equations $y = -2x + 5$ and $y = -2x$ have a solution?
Give a reason for your answer.

2.4 Do the simultaneous equations $y = x + 6$ and $y = 6 - x$ have a solution?
Give a reason for your answer.

2.5 Do the simultaneous equations $y = \frac{1}{2}x + 3$ and $y = 0.5x - 1$ have a solution?
Give a reason for your answer.

2.6 Do the simultaneous equations $x + y = 5$ and $y = -x + 2$ have a solution?
Give a reason for your answer.